



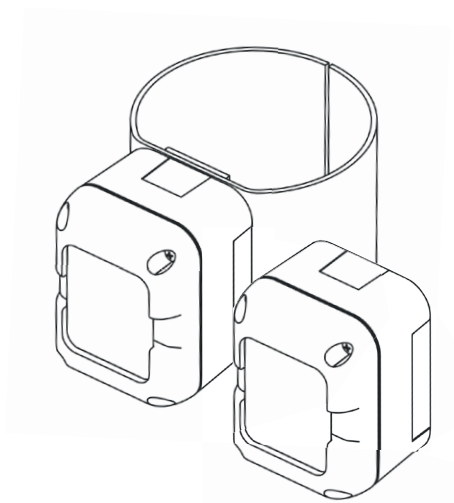
FERT SENSOR

WIRELESS SENSOR

LEARN EVERYTHING ABOUT
THE USE AND MAINTENANCE
OF YOUR SENSOR.

OPERATION MANUAL

 FertiSystem



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Presentation

FertiSystem is a Brazilian leader in the development, research, and production of fertilizer metering devices attachable to planters and seeders. Its foundation lies in the concept of the FertiSystem Auto-Lub AP NG System — a precision fertilizer metering device. It was the first of its kind on the global market, revolutionizing dosing and application systems by providing effective uniformity in the distribution of solid fertilizers, ensuring proven gains in crop productivity and profitability for farmers. Currently, FertiSystem supplies approximately 95% of Brazilian agricultural machinery and implement manufacturers, and has been operating in numerous international markets since 2008. The company has received several state and national awards for its market differentiation and remarkable technological advancements, as well as for the usability and commercial viability of its solutions. Notable recognitions include the RS Industry Distinction Award - FIERGS/CIERGS 2008, the Gerdau Best of the Land Award 2008, and the Três Porteiras Trophy from Federasul - RS in 2015 and 2016.

For over five years, the company has been engaged in research and development of new technologies and services. Since 2018, FertiSystem has been designing and delivering solutions through intelligent systems, introducing new products and digital processes such as embedded software, human-machine interfaces, electronic hardware, and sensors. These technological advancements play a strategic role in the company's future and in the sustainability of agribusiness, making a significant impact across various sectors — particularly in planting processes — through automation of fertilizer dosing and the integration of data and knowledge. The technology we bring to the field has been key to reaching our current level. Continuous evolution is now solidified through the concept and vision of digital agriculture.

Warranty Terms

The warranty for the Fert Sensor is granted exclusively to the original purchaser for a period of one (1) year from the date of purchase, following the issuance of the invoice by FertiSystem, and covers manufacturing defects that may compromise the operational performance of the product.

Conditions

A) The product is warranted against any manufacturing defects, provided it is proven that the parts and components were supplied by FertiSystem;

B) Parts and components will be replaced if the defects are verified by FertiSystem technicians, as described in item D;

C) FertiSystem guarantees the repair or replacement of parts and components free of charge, provided the warranty conditions are met. In case the warranty period has expired or the warranty has been canceled, any technical assistance provided by FertiSystem, as well as the replacement of parts and components, will be charged at current market prices;

D) All warranties for our products are considered bench warranties, meaning they do not include technician visits to the customer for installation or repairs. The Fert Sensor may only be disassembled and repaired by authorized personnel; otherwise, the warranty will be void.

E) In special situations, FertiSystem may provide its technicians free of charge for advisory purposes, in order to address specific occasional issues. However, travel, accommodation, and meal expenses related to such service shall be the responsibility of the purchaser.

Warranty Exclusion

A) Cases involving installations that do not follow proper mounting standards, misuse, abuse, negligence, or lack of proper maintenance, in disagreement with the instructions provided in the operator's manual;

B) Parts damaged due to misuse or lack of proper maintenance;

C) Damage or harm of any kind or nature, immediate or otherwise, caused by improper, unsuitable, or uncertain objects, to persons or property;

D) In particular, if the Fert Sensor shows external damage or clear evidence of misuse by the customer, these will be considered as factors responsible for the resulting damage, and the warranty shall be rendered void;

E) Any losses, costs, penalties, and damages including loss of profit; loss of planting; losses due to delays in any operations; any expenses or losses related to labor, adjustments, transmissions, or settings prescribed by manufacturers; replacement through equipment rental; debts owed by the buyer to their clients or third parties; and all other consequential damages or losses, liabilities, or negligence of the Company, its agents, employees, or subcontractors that may be alleged as a result of the use or failure in the technical performance of the Fert Sensor. In the event the buyer acts in bad faith in any replacement or return procedure, seeking undue or unfair advantage, FertiSystem shall be released from any obligations previously assumed under these conditions.

IMPORTANT: FertiSystem reserves the right to make changes to the product without prior notice.
This warranty statement supersedes all previous versions.

1. Technical Description

The Fert Sensor is a high-tech product developed to monitor the flow of granular materials in agricultural implements during planting and cultivation operations of crops such as soybeans, corn, sugarcane, among others.

The Fert Sensor is a completely wireless device. Communication with the monitor is done via radio signal, and it is powered by a high-tech non-rechargeable battery. Its electronic components are fully protected against impurities (such as dust, soil, etc.) and water, allowing the device to operate under the most demanding planting conditions without requiring maintenance.

Sensor communication is exclusively compatible with FertiSystem monitors.

1.1 Technical Specifications

Dimensions: 2.36 in x 4.33 in x 3.14 in (H x W x D)

Weight: 0.31 lb

Range: 3.2 ft with obstacles

Battery: Non-rechargeable – (Lithium 3.6 V – 4000 mAh)

Approximate lifespan of 2,500 hours of use (at 25°C)

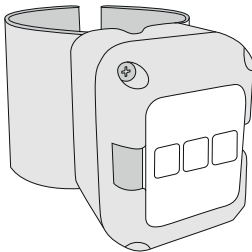
Replaceable

Communication: RF 915 MHz

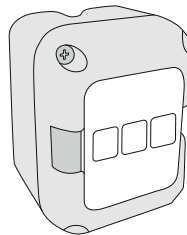
Application: Mounting on conduits from 0.98 in to 3.93 in in diameter

Operating Temperature: Up to 140°F

1.2 Components



Gravity sensor



Air Drill Sensor



Hellermann strap

Fig. 01

2. Instalation



Before starting the installation of the Fert Sensor on the implement, it is necessary to complete the addressing procedure. The step-by-step instructions for this procedure can be found in the Wireless Monitor AT200 operation manual.

1. Identify the most suitable position on the equipment for installation. Avoid locations where the sensor may come into contact with metal parts and/or moving parts of the implement. For gravity-fed systems, install the sensor using a clamp.

2. Pass the UV-resistant Hellermann strap (included with the product) through the channel on the front part of the sensor (Fig. 02).

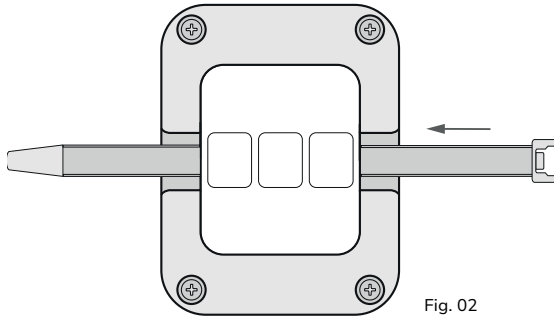


Fig. 02

3. Wrap the hose with the sensor tabs (Fig. 03). It's not a problem if the tabs overlap.

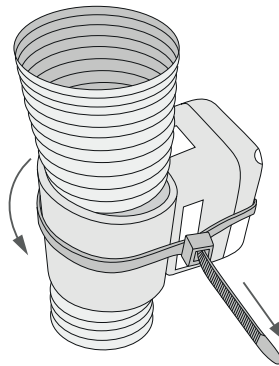


Fig. 03

4. Tighten the strap until the sensor is securely fixed, but avoid over-tightening to the point of deforming the hose and causing blockage.



Gradually tighten the strap and test the sensor's attachment by shaking the hose with movements similar to those made by the implement during operation.

5. To finish, cut off the excess strap.
6. For thin hoses and pneumatic systems, it is recommended to use the Fert Sensor Air Drill (without a clamp), as the clamp used for gravity sensors may interfere with installation and operation.

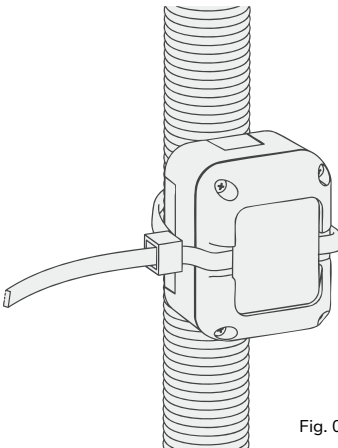


Fig. 04

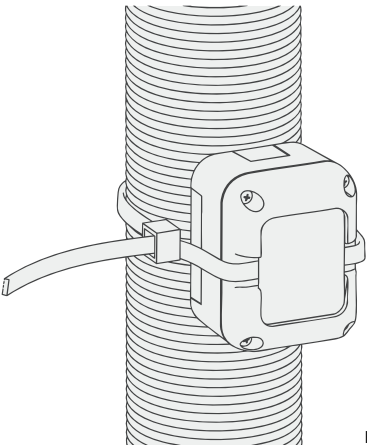


Fig. 05

2.1 Battery Replacement

After the battery reaches the end of its life, it can be replaced with a new one, thereby extending the use of the Fert Sensor. The low battery alert will be displayed on the monitor to which the sensor is connected.



There are batteries on the market with similar characteristics to the one used in the product; however, to ensure proper functioning and longevity, we recommend purchasing original batteries.

To purchase original batteries, contact the product seller directly. To replace the battery:

1. Remove the sensor from the implement by cutting the fixing strap or removing the entire hose. Battery replacement is not recommended while the sensor is installed on the implement.
2. Using a Phillips screwdriver, loosen the four fixing screws and remove the cover.
3. Pull the battery out of the casing, followed by the foam padding, to access the battery connector (Fig. 06). Note the position of the wires and terminal to facilitate connection of the new battery.

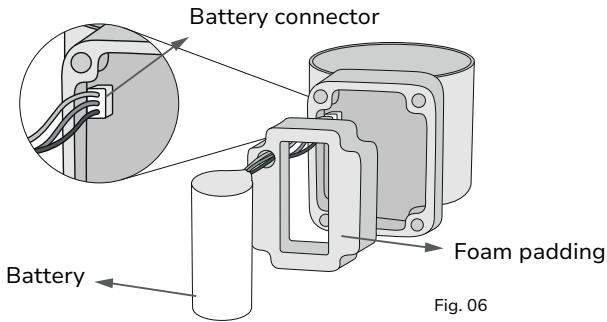


Fig. 06

4. Pull the male connector attached to the battery. Be careful not to use excessive force to avoid damaging the fixed connector (Fig. 07).



The connector has a correct assembly position. In this position, the connection is smooth and requires no force. If the connection is difficult, do not apply force; check if the connector is properly aligned.

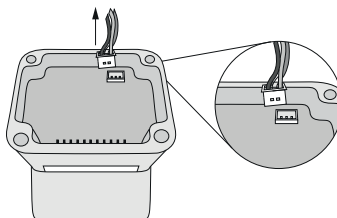
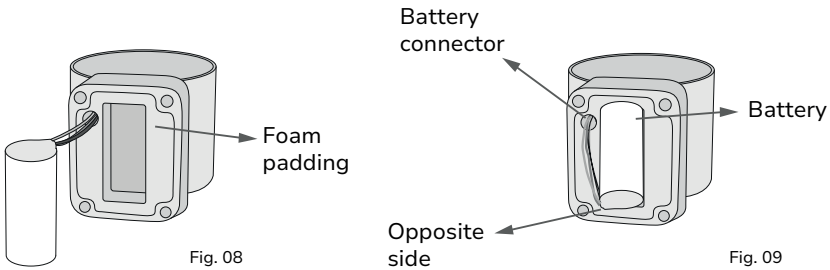


Fig. 07

5. Return the battery's protective foam and press it until it is fully seated (Fig. 08).

6. Place the battery in the designated space within the foam. Ensure that the wires coming from the battery are positioned opposite the connector (Fig. 09).



7. Replace the cover and tighten the screws until the casing is fully closed.



When replacing the battery, be careful to avoid damage to the wires, connectors, protective foam, sealing, and any cracks in the sensor housing, cover, or other components. Any damage to these parts will void the warranty.

After replacing the battery, the sensor will automatically reconnect with the monitor, retaining all its settings, including the previously configured row number.

3. Disposal

The Fert Sensor is equipped with electronic components and a lithium battery. For this reason, it should not be disposed of with regular household waste.

Disposal must be done at designated electronic waste collection points or by returning the product to the retailer, who will be responsible for proper disposal of the material.

This device contains FCC ID VVJAN1310UA-A-915

This device complies with Part 15 of the FCC Rules.

Operation is subject to the following two conditions:

- (1) This device may not cause harmful interference, and
- (2) This device must accept any interference received, including interference that cause undesired operation.

For further information, please visit www.fcc.gov

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